

Sumit Raj

CONTACT

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Portfolio: sumit966-github-io.vercel.app

SKILLS

Frontend: React.js, JavaScript, TypeScript
Backend: Node.js, FastAPI, REST APIs
Languages: JavaScript, TypeScript, Python, Java, SQL
Databases: MySQL, PostgreSQL, MongoDB
Tools and DevOps: Git, GitHub, Docker, GitHub Actions CI/CD

EDUCATION

Visvesvaraya National Institute of Technology (VNIT), Nagpur 2024 – May 2026
M.Tech, Applied AI and Machine Learning
Dr. D.Y. Patil Institute of Technology, Pune 2019 – 2023
B.E., Computer Science CGPA: 7.89/10
Certifications: Docker Mastery, Udemy (in progress); Google Cloud: Cloud Fundamentals

EXPERIENCE

Software Engineer Intern, Salesforce (Remote) Feb 2022 – May 2022

- Integrated Salesforce APIs into optimized backend workflows, improving system efficiency by approximately 25%.
- Built Python bulk data-processing pipelines, cutting manual data handling by approximately 40%.
- Wrote SQL queries with joins and aggregations to support CRM data analysis for business teams.

PROJECTS

RAG-Based Clinical Question Answering System 2025
Python, FastAPI, LangChain, ChromaDB, Docker, GitHub Actions

- Built REST APIs with FastAPI that serve LLM answers over clinical documents, reaching 92% answer relevance on a 100-question benchmark.
- Added hybrid search (BM25 + dense embeddings) with Reciprocal Rank Fusion, improving retrieval accuracy by 18%.
- Containerized with Docker and set up GitHub Actions CI/CD, keeping inference latency under 500 ms.

CI/CD Pipeline with Docker and GCP
GitHub Actions, Docker, GCP Compute Engine

- Containerized a Python web application with Docker and deployed it on GCP Compute Engine through a GitHub Actions pipeline, reducing manual deployment time by 70%.

Military Vehicle Detection and Face Authentication Security System Academic Project
Python, YOLOv8, OpenCV, Haar Cascade, Docker, GCP Cloud Run

- Built a real-time surveillance system with Haar Cascade face authentication at approximately 95% detection accuracy.
- Detected and tracked military vehicles with YOLOv8 and OpenCV, sending automated email alerts for security events.
- Containerized the YOLOv8 system with Docker and deployed it on GCP Cloud Run for serverless inference, with Cloud Logging for monitoring and alerts.

Brain Tumor Detection from MRI Scans 2024
Python, Deep Learning

- Trained a deep learning model to classify brain tumors on 3,000+ MRI scans.

Mobile Botnet Detection System 2024
Python, Scikit-learn

- Built a machine learning model to detect malicious bot communications with C&C servers by analyzing Android network traffic patterns.